HyperionDev

# Responsive Web Design

## Task

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# Introduction

You should now feel a bit more proficient at creating simple web pages using HTML and CSS. At this point, you may have noticed that the elements on the web page remain rigid and do not adapt comfortably to the dimensions of different screen sizes or accommodate accessibility, such as the use of screen readers.

A site that looks good on a PC may become squashed and cluttered on a mobile device. Similarly, a site that looks good on both a PC and a mobile device also needs to accommodate users with poor eyesight, hearing, or mobility.

Responsive design aims to build web pages that detect the visitor's screen size and orientation and change the layout accordingly.

## What is responsive web design?

Responsive design is a method of creating a web application that is able to adapt to different screen resolutions while maintaining interactivity. In other words, the exact layout will vary, but the relationship between elements and the reading order remains the same. The responsive design approach combines the following components apart from HTML and CSS:

1. Responsive images and units
2. Media queries
3. Flexible design layouts

**Take note**

It is important to note that responsive web design – **coined by Ethan Marcotte in 2010** – was a term used to describe an approach to a web design. Since then, responsive design has become the default approach with web frameworks, making it easier to design responsive sites.

# Responsive units

## The viewport

The viewport is the screen size where the web page is in view.

CSS has both absolute and relative units of measuring the viewport's dimensions. An example of an absolute unit of length is a **cm** (centimetre) or **px** (pixel). Relative units or dynamic values depend on the screen's size and resolution or the root element's font sizes.

The more common relative/responsive units are:

- `em`: relative unit based on the font size of the parent element.
- `rem`: relative unit based on the font size of the root element.
- `vh`; `vw`: percent of the viewport's height or width.
- `%`: percentage of the parent element.

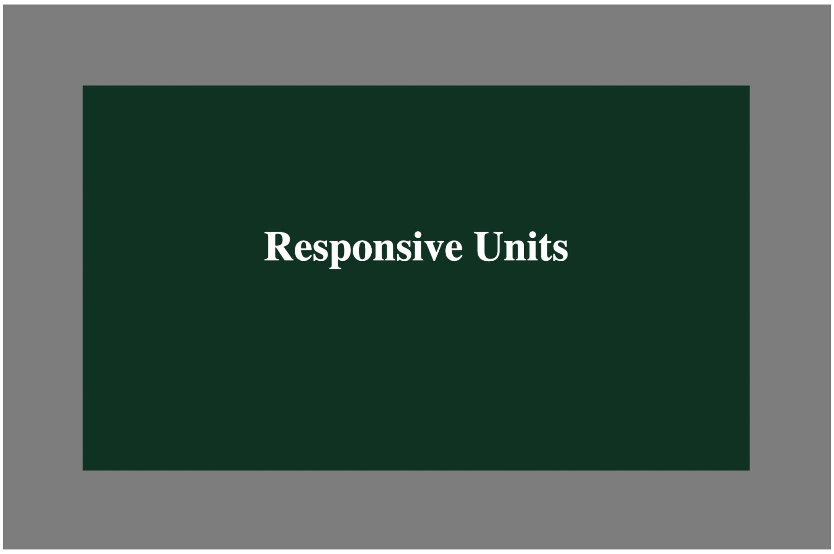
Viewport units are beneficial when an element's width and height must be specified relative to the viewport because they depend on the viewport dimensions.

```html
<!DOCTYPE html>
<html>
  <head>
    <title>Responsive Units Example</title>
    <style>
      * {
        box-sizing: border-box;
      }
      body {
        text-align: center;
        margin: 0;
        background-color: grey;
      }
      h2 {
        font-size: 5vw;
        color: white;
        padding-top: 25vh;
      }
      .centered {
        width: 80vw;
        height: 70vh;
        margin: 15vh auto;
        background: #123524;
      }
    </style>
  </head>
  <body>
    <div class="centered">
      <h2>Responsive Units</h2>
    </div>
  </body>
</html>
```

The example above uses a viewport unit to centre the element vertically. To do this, we need to know the element height. In this case, the element's height has been set to `70vh`, which means that it will take up 70% of the viewport height.

To calculate the remaining space around the element, the following formula can be used `[(100 - element height) / 2]vh`. For example, with a height of `70vh`, the formula becomes `[(100 - 70) / 2]vh`, which equals `15vh`. The result of this formula will tell us how much space will be split between the top and bottom of the element.

This approach allows for an element to be vertically centred, as shown in the image below:




## Media queries

An adaptive web design approach uses media queries. These allow us to run a series of tests (e.g., whether the user's screen is greater than a certain width or resolution) and apply CSS selectively to style the page appropriately for the user's needs.

The different media types are:

- All: default, which matches all devices.
- Print: used with printers.
- Screen: fits devices with a screen.
- Speech: fits devices with text-to-speech functionality.

The `media screen` query allows the web page to respond to different screen sizes by applying specific styles based on the screen's viewport dimensions. It helps the page automatically adjust its layout to match the size of the device being used. See the code below:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Responsive Units Example</title>
    <style>
      * {
        box-sizing: border-box;
      }
      body {
        text-align: center;
        margin: 0;
        background-color: grey;
      }
      h2 {
        font-size: 5vw;
        color: white;
        padding-top: 25vh;
      }
      .centered {
        width: 80vw;
        height: 70vh;
        margin: 15vh auto;
        background: #123524;
      }
      @media screen and (max-width: 700px) {
        .centered {
          width: 80vw;
          height: 70vh;
          margin: 15vh auto;
          background: blue;
        }
      }
    </style>
  </head>
  <body>
    <div class="centered">
      <h2>Responsive Units</h2>
    </div>
  </body>
</html>
```

We've modified the code that centres the element in the earlier example. The `media` query checks the screen (the media type), and if the width dimensions are less than or equal to 700px, it changes the background colour of the centred element to blue. The `and` keyword combines a media feature with a media type or other media features.

You can have multiple queries within a stylesheet to monitor and adjust the page layout and elements to best suit various screen sizes.

The point at which the query triggers a change in the website layout is known as a **breakpoint**. You can declare breakpoints as a specific viewport width value.

You only need to add in a breakpoint and change the design when the content starts to look bad. It is important to note that a responsive layout might appear quite different on a desktop versus a tablet or smartphone.

A mobile-first design approach would be creating a single-column layout and implementing a multiple-column layout when there is sufficient screen width to handle it. The `media` query will check for the `min-width` parameter to trigger changes as the screen size increases. If you choose this approach, you would only need to include mobile breakpoints if you optimise the design for specific models.

```css
/* Mobile-first design */
body {
  font-size: 16px;
}
.container {
  display: block; /* single-column layout */
}
@media (min-width: 768px) {
  /* When screen width is 768px or more (tablet/desktop) */
  .container {
       display: grid;
       grid-template-columns: repeat(2, 1fr); /* two-column layout */
  }
}
@media (min-width: 1024px) {
  /* When screen width is 1024px or more (desktop) */
  .container {
       grid-template-columns: repeat(3, 1fr); /* three-column layout */
  }
}
```

With a desktop-first approach, we use the parameter **`max-width`** instead of width because the styles need to be constrained below a specific viewport size with decreasing screen width. The example above depicts a desktop-first approach, but a mobile-first approach takes precedence.

```css
/* Desktop-first design */
body {
  font-size: 16px;
}
.container {
  display: grid;
  grid-template-columns: repeat(3, 1fr); /* default three-column layout */
}
@media (max-width: 1024px) {
  /* When screen width is 1024px or less (tablet) */
  .container {
       grid-template-columns: repeat(2, 1fr); /* two-column layout */
  }
}
@media (max-width: 768px) {
  /* When screen width is 768px or less (mobile) */
  .container {
       display: block; /* single-column layout */
  }
}
```

This approach ensures that the design remains adaptable and visually appealing across various devices.

# Flexible design layouts

With a flexible design, the widths of page elements will be proportional to the width of the screen or browser window. Flexible design ensures that the layout remains consistent.

# Positioning in CSS

## What is Positioning?

Positioning in the CSS allows you to control the exact placement of elements on a web page. This is achieved using the `position` property and its values:

- `static`: Default value. Elements are positioned according to the normal document flow.
- `relative`: Elements are positioned relative to their original position.
- `absolute`: Elements are positioned relative to their nearest positioned ancestor or the containing block.
- `fixed`: Elements are positioned relative to the viewport, and they stay fixed even if the page is scrolled.

## Relative and Absolute Positioning

### Relative Positioning

- Use `top`, `bottom`, `left`, and `right` properties to shift elements from their original position:

```
.element {
  position: relative;
  top: 20px;
  left: 30px;
}
```

### Absolute Positioning:

- Use `top`, `bottom`, `left`, and `right` properties to position elements relative to a containing block:

```
.container {
  position: relative;
}
.element {
  position: absolute;
  top: 10px;
  right: 10px;
}
```

### Positioning and Grid/Flexbox

Grid and Flexbox are powerful layout systems. Positioning can be used in conjunction with them to create more precise layouts:

- **Grid:** Use `position: relative` or `position: absolute` to position items within grid cells.
- **Flexbox:** Use `position: relative` or `position: absolute` to position items within flex containers.

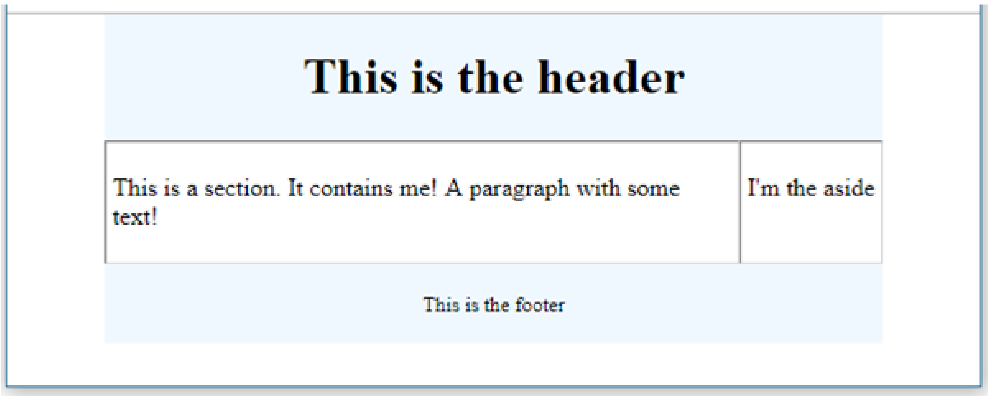
Understanding positioning is essential for creating complex and responsive layouts. By combining positioning with Grid and Flexbox (explained just below), you can achieve precise control over element placement on your web pages.

## Grid layout

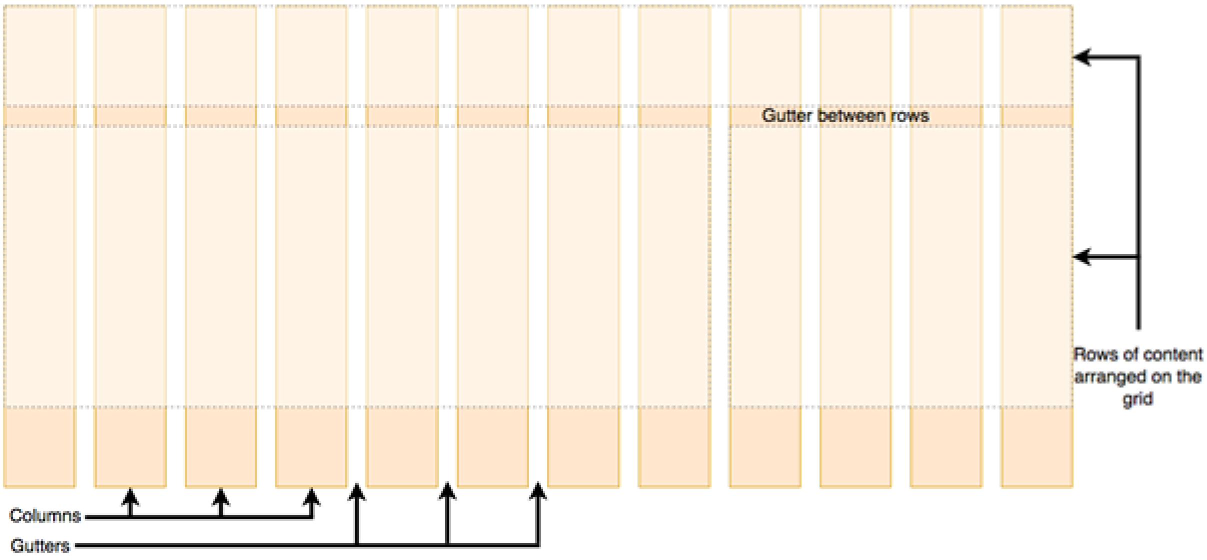
With the above knowledge on CSS positioning in mind, let's assume that we have created the following HTML:

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <title>Position Example</title>
    <link rel="stylesheet" href="example_style.css" />
  </head>
  <body>

  </body>
</html>
```

We want this HTML to be displayed in the browser, as shown in the image below:



It would be difficult to get this layout, so we can use a CSS grid template to achieve this. A CSS grid is like a table that is designed to make it easier to position elements on a web page. The grid usually contains 12 columns, as depicted below:




*CSS grid framework (Mozilla, n.d.)*

The position of an element is described in terms of which row it is in and how many columns it takes up.

To illustrate, look at the CSS rules below:

```
body {
  width: 80%;
  margin: auto;
}
#container {
  display: grid;
}
header {
  grid-column: 1/13;
  grid-row: 1;
  background-color: aliceblue;
  text-align: center;
}
section {
  grid-column: 1/9;
  grid-row: 2;
  padding: 4px;
  border: 1px inset lightgrey;
}
aside {
  grid-column: 9/13;
  grid-row: 2;
  padding: 4px;
  border: 1px inset lightgrey;
}
footer {
  grid-column: 1/13;
  grid-row: 3;
  background-color: aliceblue;
  text-align: center;
}
```

We want the `<header>` element to span the full width of the whole container/grid. Thus, we specify that we want it to start at the first line (`1`) of the grid and end at the last line (`13`) of the grid. To position the `<header>` element at the top of the grid, put it in the first row of the grid.

Place the `<section>` element in the second row of the grid. The element is eight columns wide, so the section element starts at the first line of the grid and ends at the ninth line.

Fluid grids divide the page width into equally sized columns, with content placed according to these columns. As the viewport expands, both the columns and content adjust in size.

To define a grid, we use the grid value of the `display` property. In the instance below, we have added three columns to the grid using the `grid-template-columns` property. All child elements in the grid container become grid items.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Grid Example</title>
    <style>
      .grid-container {
        display: grid;
        grid-template-columns: auto auto auto;
        background-color: purple;
      }
      .grid-item {
        background-color: plum;
        text-align: center;
        border: 1px solid black;
        padding: 10vh;
        font-size: 3vh;
      }
    </style>
  </head>
  <body>
    <div class="grid-container">
      <div class="grid-item">1</div>
      <div class="grid-item">2</div>
      <div class="grid-item">3</div>
      <div class="grid-item">4</div>
      <div class="grid-item">5</div>
      <div class="grid-item">6</div>
      <div class="grid-item">7</div>
    </div>
  </body>
</html>
```

## Flexbox layout

Flexbox is a CSS module designed to more efficiently position multiple elements, even when the size of the contents inside the container is unknown. Items in a flex container expand or shrink to the available space.

Unlike grid layouts which use columns, a Flexbox uses a single-direction layout to fill the container. A flex container expands items to fill the available free space or shrinks them to prevent overflow.

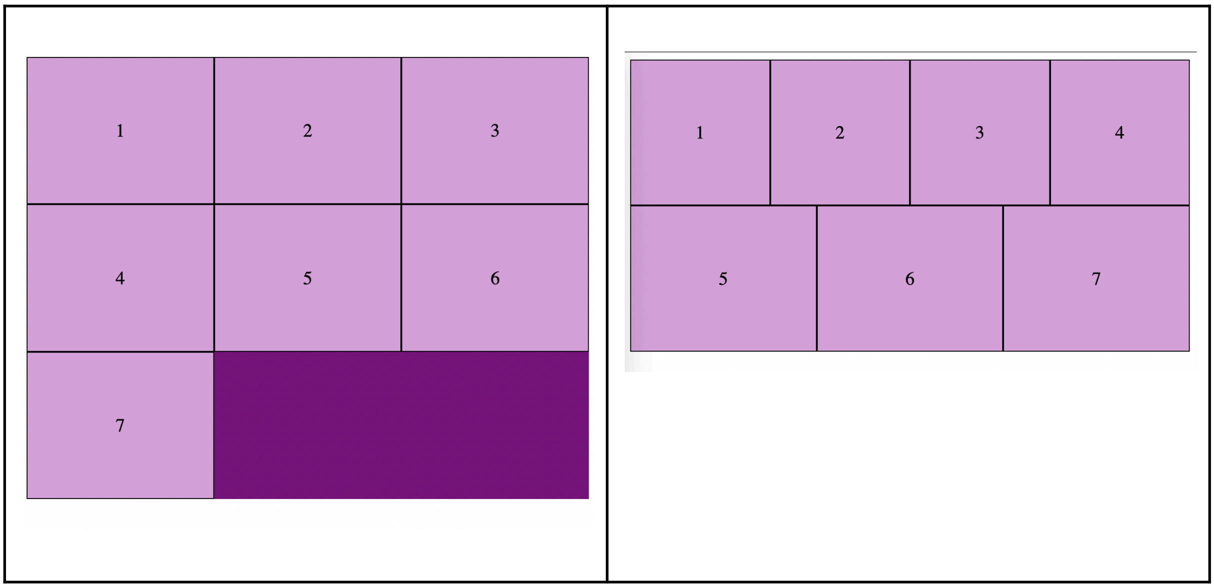
```html
<!DOCTYPE html>
<html>
  <head>
    <title>FlexBox Example</title>
    <style>
      .flex-container {
        display: flex;
        flex-flow: row wrap;
        background-color: purple;
      }

      .flex-item {
        background-color: plum;
        text-align: center;
        border: 1px solid black;
        padding: 10vh;
        font-size: 3vh;
        flex: 1;
      }
    </style>
  </head>
  <body>
    <div class="flex-container">
      <div class="flex-item">1</div>
      <div class="flex-item">2</div>
      <div class="flex-item">3</div>
      <div class="flex-item">4</div>
      <div class="flex-item">5</div>
      <div class="flex-item">6</div>
      <div class="flex-item">7</div>
    </div>
  </body>
</html>
```

In the example above, we have defined a Flexbox using the `flex` value of the `display` property. To specify the directional orientation of our Flexbox items, we can use the `flex-direction` property, which establishes the main axis/direction by row or column. By default, `flex` items will all try to fit on one line, but we can change that using the `flex-wrap` property, which allows items to wrap as needed.

The `flex-flow` property is a combination of these two, and as you can see in the example below, it allows us to specify direction and wrapping.

We also added the rule `flex: 1` to our flex item above. This is a flex property that is a unitless proportion value that dictates how much space each flex item will take up along the central axis compared to other flex items. In this case, we're giving each `.flex-item` element the same value (a value of `1`), which means they'll all take up an equal amount of the spare space left after properties like padding and margin have been set. An element with a flex value of `2` will take up twice as much of the available space.

**Grid container** **Flex container**




The image on the left is a grid container made up of seven grid items. We can see that the items are aligned with the three columns that have been declared. Take note of the available grid space, represented by a darker purple.

The image on the right is a flex container made of seven items. All items per row shared the available space equally and succeeded in filling the box regardless of the screen dimensions.

## Extra resource

Multi-column layout is a CSS module that adds support for multiple-column layout. Read here for more about **multi-column layout**.

# Responsive images

Responsive images follow the same concept as a fluid layout, using a dynamic unit to control the width or height.

One way to create a responsive image is by setting the `img width` property to a percentage value (remember the responsive unit?).

```
img {
  width: 100%;
  height: auto;
}
```

The snippet below ensures that the image is scalable but will never exceed its original using the `max-width` property.

```
img {
  max-width: 100%;
  height: auto;
}
```

The percentage unit approximates a single percentage of the viewport's width or height and ensures the image remains in proportion to the screen.

The problem with this approach is that every user has to download the full-sized image, even on mobile.

To serve different versions scaled for different devices, you need to combine the HTML `<picture>` tag with the `<source>` tag and the `srcset` attribute.

The `<source>` tag with the `srcset` attribute accepts several images with their respective widths in pixels. Used together with the `media` attribute (which accepts media queries), the image can be selected based on the viewport.

```
<picture>
  <source media="(min-width: 900px)" srcset="large-image.png" />
  <source media="(min-width: 700px)" srcset="medium-image.png" />
  <img src="small-image.png" alt="Responsive Picture" style="width: auto" />
</picture>
<p>
  Resize the browser width and the background image will change at 900px and
  700px
</p>
```

In the instance above, the page can serve two different images depending on the viewport's dimensions. The larger of the images is set as the default. Should the width dimensions of the screen be equal to or less than `700px`, the smaller image would be served.

## A quick word on accessibility

A good website works on various screen sizes, while a great one is accessible to users with assistive technologies. Responsive design can impact accessibility, such as in Firefox's high-contrast mode, which adjusts text and background for better readability. If your content relies on background images or lacks alt text for images, it could exclude users with limited vision or those using screen readers.

Here are some good practices to think about when designing any web page:

- Add descriptive alt text to all your images.
- Make use of HTML5 semantic tags, such as `<section>`, `<article>`, `<header>`, `<footer>`, `<aside>`, `<figure>`, and `<nav>`, and lean away from using non-descriptive tags like `<div>`.
- Use CSS to float your content around the page. Avoid relying on hacks such as invisible tables to position your content, as this can greatly confuse a screen reader.

Read this **web accessibility checklist** that goes into much more detail about accessibility.

### Take note

You can test your web page using multiple screen sizes with **Chrome DevTools**. You can select the mobile device or tablet of your choice to test the responsiveness of your design.

## Take note

The task below is **auto-graded**. An auto-graded task still counts towards your progression and graduation. Give it your best attempt and submit it when you are ready.

When you select “Request review”, the task is automatically complete, you do not need to wait for it to be reviewed by a mentor. You will then receive an email with a link to a model answer, as well as an overview of the approach taken to reach this answer.

Take some time to review and compare your work against the model answer. This exercise will help solidify your understanding and provide an opportunity for reflection on how to apply these concepts in future projects.

In the same email, you will also receive a link to a survey, which you can use to self-assess your submission. Once you’ve done that, feel free to progress to the next task.

## Auto-graded task

For this task, you will be recreating the **periodic table of elements**.

- The page should be responsive to changing screen dimensions until such a point where the table loses its structure, at which point it will be replaced by an image of the periodic table (making elements disappear can be done using the CSS `display` attribute).
- The table can be created using either the Grid or Flexbox layout. Both the flex and grid layouts have properties allowing one to manipulate the space between child items (i.e., `gap`, `row-gap`, and `column-gap`).
- Create at least one breakpoint that triggers a change in the styling of the web page. For example, you could change the background colour, adjust font sizes, or modify other visual elements. This ensures that the design adapts to different screen sizes, maintaining a user-friendly layout across devices.

**Important:** Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.

**Share your thoughts**

Please take some time to complete this short feedback **form** to help us ensure we provide you with the best possible learning experience.

# Reference list


Mozilla. (n.d.). *Grids*. MDN Web Docs. Retrieved June 18, 2024, from **https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout/Grids#A_CSS_Grid_grid_framework**